ENGINEERING DESIGN SUBMISSION

Bill of Materials for a Single-Motor Electric RC Aircraft

1.0–1.2 m wingspan • Conventional high-wing layout • Wireless control • Lightweight, repairable structure

Assignment	Problem 4 — Bill of Materials for an RC Aircraft
Deliverable	Submission-ready BOM, cost comparison, engineering rationale and propulsion feasibility estimate
Scope	Planning exercise only. No parts purchased. Prices are indicative online listings in INR and may change.
Design baseline	1.10 m high-wing trainer; 3S LiPo; four control servos; hand-launch capable, with optional light landing gear.

Recommended configuration: an A2212-class 1000 KV brushless outrunner on 3S LiPo with a 30 A ESC, beginning with a conservative 9×4.5 or 10×4.5 two-blade propeller only after checking the motor/propeller current limit. A high-wing foam-board airframe with a carbon or wood spar is inexpensive and can be repaired after landing damage.

Prepared as an engineering estimate • India market • 26 October 2026

1. Executive summary

This BOM specifies a functional, single-motor fixed-wing aircraft with a 1.0–1.2 m wingspan. A conventional high-wing configuration is selected because it is easier to trim and repair than a highly manoeuvrable low-wing design. The primary controls are ailerons, elevator and rudder, operated by four servos (two ailerons, one elevator and one rudder).

The preferred balance is the **budget/safe configuration**, provided the airframe is kept light and the propulsion system is tested before flight. The largest cost is the radio set; borrowing a compatible radio and LiPo charger from a club can reduce the outlay substantially without compromising airworthiness. The motor/propeller combination is a starting point, not a guarantee: exact thrust and current depend on the motor batch, propeller, battery state and installation.

Important limitation: vendor pages sometimes list drone-oriented propeller pairs or optimistic thrust figures. Use a correctly sized, single two-blade aircraft propeller, confirm rotation and clearance, and measure current/static thrust safely before flying. Do not run a propeller indoors or hold the aircraft by hand during a powered test.

2. Design assumptions and requirements

Parameter	Design assumption / rationale
Wingspan	1.10 m nominal, within the required 1.0–1.2 m range.
Configuration	Conventional high wing; tractor motor at nose; conventional tail.
Mission	Stable sport/trainer flight and ground or hand launch; not an aerobatic competition model.
Control surfaces	Ailerons + elevator + rudder; four servos. Flaps are omitted to save mass and channels.
Power	3S LiPo (11.1 V nominal, 12.6 V fully charged); brushless outrunner and brushless ESC.
Structure	5 mm foam board with a carbon/wood spar, plywood firewall and local reinforcement.
Target mass	Approximately 700–900 g all-up; 800 g nominal estimate used in the bonus analysis.
Price basis	Indicative Indian online retail prices, observed October 2026; excludes exact shipping/tax differences unless listing states otherwise.
Safety	Use a LiPo balance charger, correct failsafe, secure battery, propeller safety zone and supervised first flights.

3. Complete bill of materials and options

Every major item has two viable options. Option prices are planning estimates per quantity shown; the total columns below are not a quote. Links are sources for example products/specifications, not endorsements. Check stock, shipping, return policy and exact SKU before any later purchase.

3.1 Main propulsion, radio, controls and structure

Component / qty	Recommendation 1	Recommendation 2	Compatibility checks	Est. ■ (opt. 1 / 2)
Motor + ESC + prop set Qty 1	A2212 1000 KV bundle with 30 A ESC and 10×4.5 propeller pair. Approx. 150 W class, 2–3S.	A2212 1000 KV motor + separate 30 A ESC + single 9×4.5/10×4.5 prop chosen after test.	Start with conservative prop. Measure full-throttle current; do not exceed motor/ESC ratings. Vendor thrust claims vary.	1,000 / 1,400
Battery Qty 1	3S 2200 mAh 25C LiPo, XT60, approx. 165–180 g.	3S 1800–2200 mAh 25–35C LiPo from a reputable RC brand.	3S only if motor/ESC support it; 25C × 2.2 Ah = theoretical 55 A, but actual battery limits and voltage sag matter.	1,250 / 1,900
Transmitter + receiver Qty 1 set	FlySky FS-i6 + FS-iA6B; 6 channels, 2.4 GHz AFHDS 2A.	FS-i6 with a compatible FS-iA6/FS-iA6B receiver bundle.	Transmitter and receiver must use the same RF protocol and bind. Confirm receiver PWM outputs and voltage range.	5,400 / 6,200
Servos Qty 4	SG90 9 g analog micro servos.	Metal-gear 9–12 g digital servos.	Power from receiver/BEC at supported voltage. Check stall current and torque; use separate BEC if total servo load exceeds ESC BEC.	500 / 1,000
Airframe material Qty 1 lot	5 mm foam board, 3–4 sheets.	EPP/EPO foam or purpose-made foam airframe/kit.	Foam board is cheap and repairable but paper skins can absorb moisture; EPP is more impact tolerant and often costlier.	500 / 1,400
Wing spar / reinforcement Qty 1 lot	Carbon tube/strip, 6–8 mm class.	6 mm spruce/pine strip or bamboo reinforcement.	Spar near 25–30% chord; carry it continuously through the wing centre and reinforce joiner area.	250 / 650
Firewall + longerons Qty 1 lot	3 mm plywood firewall plus light balsa/ply strips.	Foam-board doublers with 3 mm birch plywood firewall.	Firewall should transfer motor loads to fuselage; use bolts and backing plate rather than glue alone.	200 / 450

3. Complete BOM (continued)

3.2 Linkages, landing system, electrical and assembly materials

Component / qty	Recommendation 1	Recommendation 2	Purpose / checks	Est. ■ (opt. 1 / 2)
Control horns, pushrods, clevises Qty 1 lot	Nylon horns, 1 mm steel pushrods and clevises.	2 mm carbon pushrods and adjustable linkages.	Four runs: two ailerons, elevator, rudder. No binding; clevises secured; neutral surfaces at servo centre.	250 / 500
Landing gear / wheels Qty 1 set	Two main wheels, wire/ply gear, optional tail skid.	Hand-launch belly lander with foam skid/reinforcement.	Ground takeoff requires strong gear mounts; hand launch saves mass but needs a safe launch and landing area.	250 / 650
Battery connector + wiring Qty 1 lot	XT60 pigtail, 16 AWG silicone wire, heat-shrink.	Reuse correctly rated pre-fitted XT60 leads; 14–16 AWG main leads.	Polarity must match; insulate solder joints; wires secured away from propeller and control linkages.	250 / 500
Motor bullets / prop adapter Qty 1 lot	3.5 mm bullet connectors and adapter for 3.17 mm shaft.	Matched motor/ESC connectors and compatible collet adapter.	Check shaft diameter, prop hub bore, adapter seating and prop rotation. Balance propeller before use.	100 / 250
Adhesives + assembly consumables Qty 1 lot	Foam-safe CA or UHU POR, hot glue, epoxy at firewall.	Foam-safe contact adhesive plus light epoxy at high-load joints.	Test on offcut first; solvent adhesives can dissolve foam. Keep adhesive weight under control.	300 / 650
Tape, hinges, hook-and-loop, cable ties Qty 1 lot	Reinforced packing tape, hinge tape, Velcro battery strap.	Glass-fibre tape and reusable battery strap.	Hinges must move freely; battery must not shift under acceleration or impact.	200 / 450
Battery charger Qty 1	B6-class balance charger supporting 3S LiPo.	Quality AC balance charger with cell voltage display.	A balance charger is mandatory for LiPo charging; follow manufacturer charge rate and never leave charging unattended.	1,000 / 2,200
Transmitter batteries Qty 1 set	Four AA alkaline cells.	Four rechargeable NiMH AA cells plus charger.	Verify the transmitter's allowed input voltage; check battery warning before each session.	100 / 450
Wattmeter / thrust test Qty 1 (optional)	Inline wattmeter rated 60–100 A.	Borrow a wattmeter and guarded thrust stand/scale from the lab or club.	Measure voltage, current, watts and static thrust using the intended propeller; test behind a safety barrier.	700 / 1,800

Included in totals: the chosen build should contain one item/lot for each row, except the optional wattmeter may be borrowed. A motor/ESC/prop bundle may include drone-style counter-rotating props; use only the appropriate single propeller for the aircraft and retain spares separately.

4. Budget estimate and configurations

Costs are approximate and rounded to the nearest ■50–■100 where appropriate. They represent indicative listings researched for this assignment, not guaranteed checkout prices. Shipping, regional availability and taxes can change totals.

Budget / safe

Cost line	Estimate (■)
Motor set	1,000
Battery	1,250
Radio	5,400
Servos	500
Foam	500
Spars & firewall	450
Linkages	250
Landing / skid	250
Wiring & connectors	350
Adhesives / tape	450
Charger	1,000
AA cells	100
Contingency / shipping	800
Total	12,300

Performance / durable

Cost line	Estimate (■)
Motor set	1,400
Battery	1,900
Radio	6,200
Servos	1,000
Foam / EPP	1,400
Spars & firewall	1,100
Linkages	500
Landing / skid	650
Wiring & connectors	650
Adhesives / tape	1,100
Charger	2,200
AA cells	450
Wattmeter / test	1,800
Contingency / shipping	1,200
Total	21,550

Interpretation

The budget-oriented build is estimated at ■**12,050** (with a contingency allowance). The performance-oriented/durable build is estimated at ■**20,250**. A compatible radio and charger are high-cost reusable items: borrowing either can reduce the first-build cash cost by several thousand rupees. Do not save money by using a damaged LiPo, undersized ESC, unverified receiver, weak prop adapter or poorly secured battery.

Recommended for competition: budget/safe, upgraded only where testing reveals a need. It is sufficient for a light trainer if the propulsion system passes current/thrust checks and the structure is stiff. Put available funds into a reliable radio link, quality battery/charger, spar and repeatable pre-flight testing rather than cosmetic finish.

5. Engineering justification

5.1 Motor–propeller match and thrust check

A 1000 KV outrunner on a 3S pack has a theoretical no-load speed of approximately 1000 rpm/V × 11.1 V = 11,100 rpm at nominal voltage (about 12,600 rpm at 12.6 V full charge). Loaded RPM will be lower. A 10×4.5 propeller can generate useful static thrust at modest airspeed, but it may draw too much current for a particular motor. The exact propeller must therefore be selected from a motor manufacturer's thrust/current table or tested using a wattmeter and thrust stand. If current exceeds the motor's continuous rating, step down in diameter and/or pitch.

Test sequence: (1) secure the motor to a rigid test stand behind a barrier; (2) use a correctly rated wattmeter and fully charged, healthy 3S battery; (3) keep people and loose objects outside the propeller plane; (4) increase throttle gradually and record voltage, current, watts and thrust; (5) stop immediately if vibration, unusual sound, overheating or current limit is observed. Never hold the aircraft by hand for a full-throttle test.

A useful initial screening target for a stable trainer is static thrust at least 0.5 times all-up mass for acceptable takeoff/climb margin; higher is useful but must not compromise current, temperature or structural limits. A thrust-to-weight ratio is only a screening measure; wing loading, airspeed, drag and takeoff surface also determine performance.

5.2 ESC rating and electrical margin

Use a 30 A ESC for the selected motor class, assuming measured full-throttle current remains at or below approximately 20–24 A in the final propeller/battery configuration. This preserves roughly 20–33% current headroom. The ESC must support 3S LiPo voltage and have a BEC output compatible with the receiver and servos. A listing for the A2212 class may quote lower motor current limits than the ESC's 30 A rating; the motor rating, not the ESC rating, is the limiting constraint. If measured current is above the motor's limit, reduce prop load rather than relying on the ESC's larger rating.

BEC current also matters: four servos can draw short current peaks, especially when loaded. Check the ESC BEC's continuous rating and servo stall-current estimates. If the BEC cannot safely support the combined receiver/servo load, use a suitable independent BEC and isolate wiring as directed by its manufacturer.

5.3 Battery selection and trade-offs

A 3S 2200 mAh pack is a common compromise between power, endurance and mass. Theoretical energy is 11.1 V × 2.2 Ah = 24.4 Wh. At a planning average of 10–15 A, ideal endurance would be 8.8–13.2 minutes if the full capacity were used; preserving a 20–25% reserve and allowing for throttle variation gives a rough practical estimate of about 7–10 minutes. This is not a flight-time guarantee. A larger battery adds energy but increases wing loading and landing energy; a smaller battery reduces mass but may shorten flights and experience more voltage sag.

The stated 25C rating gives a theoretical discharge figure of 2.2 Ah × 25 = 55 A. This is not proof that the pack can safely deliver 55 A continuously: manufacturer quality, temperature, connector condition and voltage sag must be considered. Use a balance charger, inspect for swelling or damage, avoid deep discharge, and store at the manufacturer's recommended storage voltage in a fire-resistant location.

5.4 Servos and control surfaces

Four servos are selected: two aileron servos (one per wing panel), one elevator servo and one rudder servo. This provides conventional three-axis control and allows aileron differential or trim adjustments if the transmitter supports them. A two-servo elevon arrangement would suit a flying wing, but that is not the selected configuration. Servo torque must be adequate for surface area, hinge friction and airspeed; control throws should follow the design and be checked at low speed before flight.

5.5 Structure: strength, mass, cost and repairability

Five-millimetre foam board is inexpensive, easy to cut and straightforward to repair with foam-safe glue and tape. Its disadvantages are paper-skin moisture absorption, local crushing and lower torsional stiffness than a purpose-built composite airframe. A carbon tube/strip spar gives high bending stiffness for low mass; spruce or bamboo is cheaper and easier to obtain but generally heavier for comparable stiffness. Use plywood at the firewall, wing centre and landing-gear mounts, where concentrated loads occur. Avoid excessive hot glue and decorative parts because they increase mass without adding strength.

5.6 Electrical and mechanical compatibility

• Radio: transmitter and receiver must share the same protocol (e.g. AFHDS 2A) and bind successfully. A 2.4 GHz label alone does not guarantee compatibility.

• Receiver: verify supply voltage range, PWM channel outputs, channel order and failsafe behaviour. Set loss-of-signal failsafe to motor off and safe control positions.

• ESC: match battery cell count, connector polarity and current rating. Confirm whether its BEC powers the receiver; do not connect two BEC outputs together unless designed for it.

• Motor/prop: match shaft diameter, prop adapter, propeller bore, rotation, clearance and mounting screws. Use a single aircraft propeller of the correct orientation; balance it.

• Servo linkages: check travel direction, neutral position, full movement without binding, secure horns and no servo buzzing at endpoints.

• Airframe: check wing incidence, straightness, centre of gravity (CG), spar continuity, battery restraint, motor thrust line and access to wiring.

• Before every flight: transmitter on first, then aircraft; range check; control direction check; failsafe check; battery check; propeller/mount inspection; clear launch/landing zone.

6. Bonus: takeoff mass and propulsion sufficiency

The estimate below is a design budget, not a measured result. It assumes a simple 1.10 m foam-board high-wing trainer, 3S 2200 mAh battery, four micro servos and a lightweight hand-launch arrangement.

Subsystem	Estimated mass (g)	Basis
Foam airframe, tail and skins	230	Cut foam board, local doublers and tape
Spar, firewall and reinforcement	55	Carbon/wood spar plus plywood motor mount
Motor + ESC + prop + adapter	120	A2212-class set, ESC, prop and mounting hardware
3S 2200 mAh LiPo	175	Typical pack class; actual SKU varies
Four servos	36	4 × approx. 9 g, excluding minor leads
Radio receiver	10	Receiver only; transmitter excluded from aircraft mass
Pushrods, horns and hinges	45	Four control runs and hinges
Wiring, connectors, adhesive and tape	55	Conservative allowance
Landing gear / skid	45	Optional light gear; less if belly lander
TOTAL ESTIMATED AUW	771	Rounded design estimate; target range 700–900 g

6.1 Thrust-to-weight screening

Assume an all-up mass of 0.771 kg. Weight force is $W = m\,g = 0.771 \times 9.81 \approx 7.56$ N. If a guarded bench test measures 0.60–0.80 kgf static thrust (approximately 5.9–7.8 N), thrust-to-weight ratio would be 0.60/0.771 = 0.78 to 0.80/0.771 = 1.04. This would generally provide useful climb margin for a small trainer, but the lower end leaves less margin for a heavy airframe, poor propeller efficiency or a grass takeoff. A more conservative acceptance criterion is ≥0.5:1 static thrust-to-weight, with a preference for approximately 0.7:1 or higher if current and temperature limits permit.

Important uncertainty: public listings for A2212 motors report different thrust figures (roughly 0.8–1.2 kg with a 10×4.5 propeller), demonstrating variation between vendors and test conditions. Do not use those values as guaranteed performance. The aircraft is considered propulsion-sufficient only after the actual motor/prop/battery setup demonstrates safe current, acceptable temperature and sufficient static thrust.

6.2 Wing loading cross-check

For a preliminary rectangular wing of 1.10 m span and 0.20 m mean chord, wing area is approximately 0.22 m² (22 dm²). At 0.771 kg, wing loading is 771 g / 22 dm² ≈ 35 g/dm². This is plausible for a small foam trainer but not especially slow; a larger chord/area or lower mass would improve low-speed handling. Final stall speed cannot be responsibly stated without airfoil lift data and actual finished geometry.

7. Source links and price evidence

The links below support example product specifications and indicative prices. Prices and stock are snapshots from online listings accessed/researched for this assignment; they may change. All figures in the BOM are estimates rounded for planning and should be rechecked before purchase.

- S1 — SunRobotics: A2212 1000 KV motor + 30 A ESC + 1045 prop set; listing gives 2–3S compatibility, 30 A ESC and BEC details.
- S2 — Online Electronics Shop: A2212 1000 KV motor listing; example motor specs and claimed static thrust values (vendor claims differ).
- S3 — Electronic Spices: 3S 2200 mAh 25C LiPo, 11.1 V, XT60; example listing.
- S4 — RC Mumbai: Gens Ace 3S 2200 mAh 25C battery specifications; listing showed out of stock at time of research.
- S5 — SunRobotics: FlySky FS-i6 + FS-iA6B; 6 channels, 2.4 GHz and receiver voltage range.
- S6 — FlyRobo: FlySky FS-i6 6-channel radio bundle; example price and specification.
- S7 — Xbotics: SG90 9 g servo, example listing and price.
- S8 — RotorYard: airframe category and foam-board/EPP airframe price context.
- S9 — Example local-material estimate for carbon tube/strip, spruce/bamboo and plywood. Exact SKU is design-dependent; use supplier quotations before purchasing.
- S10 — Xbotics: XT60 connector, 16 AWG silicone wire, heat-shrink and bullet connector examples.
- S11 — RC Benchmark propeller test report: describes thrust/current test stand, safety cage, calibration and measurement procedure.
- S12 — BMFA Flight Club: propeller sizing and static thrust/weight discussion; useful background, not a substitute for manufacturer data.

8. Final recommendation

Build the 1.10 m high-wing foam-board trainer with a continuous spar, plywood firewall, 3S 2200 mAh LiPo, 1000 KV A2212-class motor, 30 A ESC, conservative two-blade propeller and four servos. Use the budget configuration, but treat the motor/propeller combination as provisional until measured. Confirm the exact motor continuous-current rating, ESC BEC capacity, radio binding/failsafe, CG, structural stiffness and thrust/current on a guarded test stand before the first flight. For a competition aircraft, repeatability and safe, verified performance are more valuable than a low-cost part with unknown ratings.

Submission note: This is a planning-level BOM. It does not claim that all listed alternatives have been independently flight-tested together. Exact airframe geometry, finished CG and final propeller should be validated during prototyping.